BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 117 · File Handling

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the scope of a local variable or constant?

Q2. What happens in the fetch stage?
A. Get the next instruction
B. Work out its meaning
C. Carry out the instruction
D. Save a backup copy

Q3. “Execution only copies an instruction into memory without performing it.” Correct this. Explain why.

Q4. “The Control Unit stores your documents forever.” Correct this. Explain its role when the CPU runs an instruction.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Open — A file is opened before reading or writing through a program.

Read — Input from an existing file can be processed by a program.

Write — Output can be saved to a file, with appropriate mode and permissions.

Close — Closing releases the file resource and helps ensure changes are saved.

New learning

Persistent storage: A text file keeps data after the program ends. A filename identifies a file, but does not itself hold the file's contents.

Open and read: OCR ERL stores an opened file in a variable: f = open('pupils.csv'); f.readLine() returns the next line. This file has a header.

Write and close: f.writeLine(text) writes a line to an opened output file; f.close() releases the file handle.

CSV and fields: In this teaching example, each line separates ID, Name and Mark with commas. Splitting CSV fields into an array is an optional extension, not a separate compulsory OCR requirement.

Worked example

The sample file pupils.csv contains a header followed by 101,Ada,72; 102,Ben,43; 103,Mia,81. Trace two readLine() calls and state the lines returned.

1. Open pupils.csv with file = open('pupils.csv'); save the opened file in a variable.

2. First readLine() returns the header: ID,Name,Mark. The next readLine() returns 101,Ada,72.

3. Each further readLine() advances to the next line. Use endOfFile() when reading the whole file.

4. Close the file. Reading a line returns text; it does not automatically turn comma-separated fields into a 2D array.

We Do · Practise with guidance

Guided 1

A program creates a separate results.txt containing the lines 'Ada,72' and 'Mia,81'. Outline how it writes and then closes the file. [4 marks]

Worked method: Close the file when finished; identify the order open → write → write → close.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the instruction that prints each successive line in the supplied file.

```
myFile = open("sizes.txt")
while NOT myFile.endOfFile()
    print(...)
endwhile
myFile.close()
```

write

Question 1

Name the four basic file-handling operations required by OCR J277. [4 marks]

Question 2

Given pupils.csv with a header followed by 101,Ada,72 and 102,Ben,43, state what the second call to readLine() returns after opening the file. [2 marks]

Question 3

Explain why the file should be closed after reading its final line. [2 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A file should normally be closed after use.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A local variable can be used only within its own subprogram or local scope.

Q2. A — Get the next instruction

Q3. The operation specified by the decoded instruction is carried out.

Q4. The Control Unit decodes instructions and controls CPU operations. The ALU processes data; registers store it briefly.

I Do / We Do pages · Guided practice

Q1. Close the file when finished; identify the order open → write → write → close.

You Do pages · Code First

```
myFile = open("sizes.txt")
while NOT myFile.endOfFile()
    print(myFile.readLine())
endwhile
myFile.close()
```

Expected result: 4 6 10 (if sizes.txt contains these three lines)

You Do pages · Independent questions

Q1. Open [1], Read [1], Write [1], Close [1].

Q2. 101,Ada,72 [1]; because the first readLine() already returned the header [1].

Q3. Releases the opened file resource/handle [1] and properly completes the file access [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: